Monetary Policy

ECON 101 - Macroeconomics

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Chapter Outline

- **11.1** What Is Monetary Policy?
- **11.2** The Money Market and the Bank of Canada's Choice of Monetary Policy Targets
- **11.3** Monetary Policy and Economic Activity
- **11.5** A Closer Look at the Bank of Canada's Setting of Monetary Policy Targets

11.1 Learning Objective

Goal

Define monetary policy and describe the Bank of Canada's monetary policy goals.

- Build on the previous chapter on money and banking.
- Understand **what** the Bank of Canada (BoC) is trying to achieve.
- See **why** monetary policy matters for inflation, unemployment, and growth.

What Is the Role of the Bank of Canada?

- Created in **1934**.
- Main responsibility:
  - conduct monetary policy "in the best interests of the economic life of the nation."
- Since World War II:
  - the BoC has carried out an **active** monetary policy.
- **Monetary policy**:
  - actions the Bank of Canada takes to manage the money supply and interest rates
  - in order to pursue macroeconomic policy goals.

The Goals of Monetary Policy

- The Bank of Canada pursues four main monetary policy goals:

Goal	Meaning / Why it matters
Price stability	Keep inflation low and stable so money remains a reliable unit of account and store of value.
High employment	Keep the economy close to full employment, avoiding unnecessarily high unemployment.
Stability of financial markets and institutions	Ensure banks and financial markets function smoothly, avoiding crises that disrupt credit.
Economic growth	Support long-run growth in real GDP by providing a stable monetary environment.

Why These Goals Are Connected

- In the long run:
  - healthy growth requires price stability and a stable financial system.
- In the short run:
  - reducing inflation too aggressively can increase unemployment,
  - easing too much to reduce unemployment can fuel inflation.
- Monetary policy involves trade-offs, especially in the short run.

11.2 Learning Objective

Goal

Describe the Bank of Canada's monetary policy targets and explain how expansionary and contractionary policies affect the interest rate.

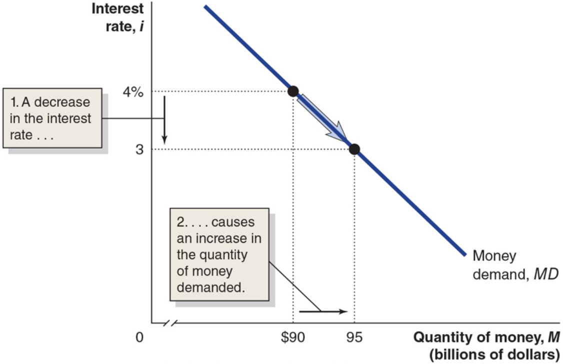
- The BoC has **two main tools**:
  - open market operations,
  - lending to financial institutions.
- These tools influence **monetary policy targets**:
  - the money supply,
  - the short-term interest rate (primary target).

Monetary Policy Tools and Targets

- **Open market operations**:
  - the BoC buys or sells government securities.
- **Lending to financial institutions**:
  - provides short-term funds to banks (standing liquidity facilities).
- The BoC cannot directly control:
  - the inflation rate,
  - real GDP or unemployment.
- Instead, it targets:
  - the money supply, and especially
  - a short-term nominal interest rate.

Figure 11.2 The Demand for Money

The money demand curve slopes downward because lower interest rates cause households and firms to switch from financial assets such as Treasury bills and Canada bonds to money. All other things being equal, a fall in the interest rate from 4 percent to 3 percent will increase the quantity of money demanded from $90 billion to $95 billion. An increase in the interest rate will decrease the quantity of money demanded.

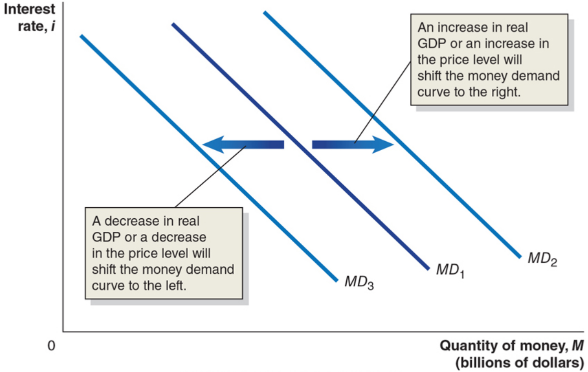


The Demand for Money

- The BoC's two monetary policy targets are **linked**:
  - higher interest rates $\Rightarrow$ lower quantity of money demanded.
- Why?
  - When interest rates are high, alternatives to holding money (e.g., bonds) look attractive.
  - The opportunity cost of holding money is higher.
- Result:
  - as the nominal interest rate rises, households and firms hold **less** money and more interest-bearing assets.

Figure 11.3 Shifts in the Money Demand Curve

Changes in real GDP, the price level, or technology cause the money demand curve to shift. An increase in real GDP or an increase in the price level will cause the money demand curve to shift to the right, from MD_1 to MD_2. A decrease in real GDP, a decrease in the price level, or more widespread use of technology like mobile wallets will cause the money demand curve to shift to the left, from MD_1 to MD_3.



Shifts in Money Demand

- Money demand depends on the **need to hold money** to pay for goods and services.
- If more transactions are taking place:
  - higher real GDP,
  - higher price level (more dollars needed per transaction),
- then the demand for money is **higher**.
- Conversely:
  - decreases in real GDP or the price level reduce money demand.

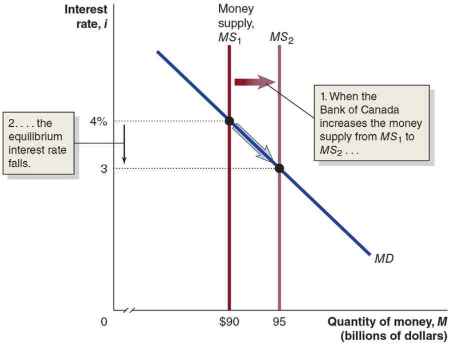
How the BoC Manages the Money Supply

- To **increase** the money supply:
  - the BoC buys government securities,
  - the sellers deposit the proceeds into bank accounts,
  - banks make new loans, and the money supply **expands**.
- To **decrease** the money supply:
  - the BoC sells government securities,
  - buyers pay from their bank deposits,
  - bank reserves fall, loans contract, and the money supply **shrinks**.

Figure 11.4 – Increasing the Money Supply

Figure 11.4 The Effect on the Interest Rate When the Bank of Canada Increases the Money Supply

When the Bank of Canada increases the money supply, households and firms will initially hold more money than they want, relative to other financial assets. Households and firms use the money they don't want to hold to buy short-term financial assets, such as Treasury bills, and make deposits in interest-paying bank accounts. This increase in demand allows banks and sellers of Treasury bills and similar securities to offer a lower interest rate. Eventually, the interest rate will fall enough that households and firms will be willing to hold the additional money the Bank of Canada has created. In the figure, an increase in the money supply from \$90 billion to \$95 billion causes the money supply curve to shift to the right, from MS_1 to MS_2, and causes the equilibrium interest rate to fall from 4 percent to 3 percent.



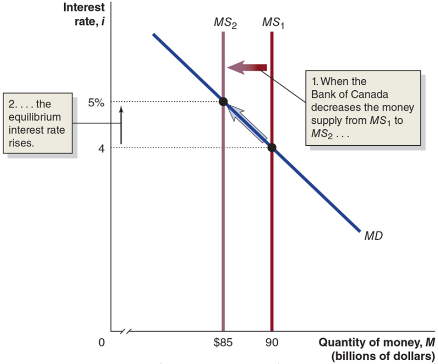
Effect of Increasing the Money Supply on Interest Rates

- Assume, for simplicity, that the BoC can fully control the money supply:
  - the money supply curve is a **vertical** line.
- Equilibrium occurs where money demand intersects money supply.
- When the BoC increases the money supply:
  - the vertical money supply curve shifts right,
  - households and firms now hold more money than they want,
  - they buy bonds and other assets,
  - bond prices rise, and the short-term interest rate **falls**.

Figure 11.5 – Decreasing the Money Supply

Figure 11.5 The Effect on the Interest Rate When the Bank of Canada Decreases the Money Supply

When the Bank of Canada decreases the money supply, households and firms will initially hold less money than they want, relative to other financial assets. Households and firms will sell Treasury bills and other short-term financial assets and withdraw money from interest-paying bank accounts. These actions will increase interest rates. Eventually, interest rates will rise to the point at which households and firms will be willing to hold the smaller amount of money that results from the Bank of Canada's actions. In the figure, a reduction in the money supply from \$90 billion to \$85 billion causes the money supply curve to shift to the left, from MS_1 to MS_2, and causes the equilibrium interest rate to rise from 4 percent to 5 percent.



Effect of Decreasing the Money Supply on Interest Rates

- When the BoC **lowers** the money supply (e.g., by selling Canada bonds):
  - households and firms use money to buy the bonds,
  - they are left with less money than they desire relative to other assets.
- To attract and keep depositors, banks:
  - offer higher interest rates on deposits.
- Result:
  - the short-term nominal interest rate **rises**.

A Tale of Two Interest Rates

- We now have two models of interest rates:
  1. **Loanable funds model** (Chapter 6)
     - determines the long-term real interest rate,
     - relevant for firms' capital investment and households building new homes.
  2. **Money market model** (this chapter)
     - determines the short-term nominal interest rate,
     - most relevant for the BoC, since changes in the money supply directly affect this rate.
- Usually:
  - long-term and short-term rates move together.

Choosing a Monetary Policy Target

- The BoC can **target** either:
  - a particular level of the money supply, or
  - a particular short-term nominal interest rate.
- In practice, it focuses on the **interest rate**.
- There are many interest rates; the key one is the:

Overnight interest rate

The interest rate banks charge each other for overnight loans.

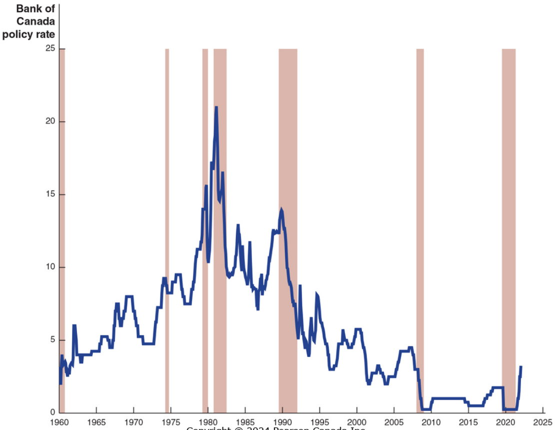
- The BoC announces its **target overnight rate** on eight fixed dates each year and uses open market operations to keep the actual overnight rate close to target.

Figure 11.6 – Overnight Rate Targeting, 1960–2022

Figure 11.6 Overnight Interest Rate Targeting, July 1960–September 2022

The Bank of Canada does not actually set the overnight interest rate. However, the Bank's ability to increase or decrease bank reserves quickly through open market buyback operations keeps the overnight interest rate close to the Bank's target rate. The Bank has pushed the overnight interest rate up and down as it has attempted to achieve its policy goals of price stability. The shaded areas represent periods of recession.

SOURCE: Bank for International Settlements, www.bis.org.



Overnight Rate and the Business Cycle

- Over the years, the BoC has **raised and lowered** the overnight rate to achieve its goals.
- During:
  - the 2007–2009 financial crisis,
  - and the COVID-19 recession in 2020,
- the BoC **cut** the target overnight rate several times.
- A change in the overnight rate has:
  - a larger effect on short-term interest rates,
  - but still influences longer-term rates and broader financial conditions.

11.3 Learning Objective

Goal

Use aggregate demand and aggregate supply graphs to show the effects of monetary policy on real GDP and the price level.

- The BoC's ability to affect real GDP depends on its ability to influence **long-term real interest rates**.
- It uses the **overnight rate** (a short-term nominal rate) as a tool.
- We assume here that changes in the overnight rate can affect long-term real interest rates.

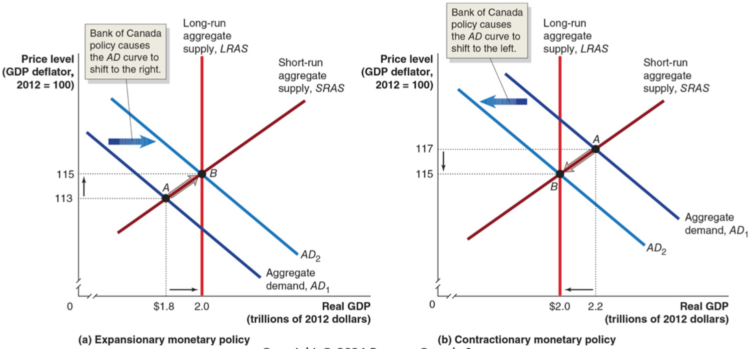
How Interest Rates Affect Aggregate Demand

- Interest rates affect:
  1. **Consumption**
     - Lower rates encourage buying on credit, especially durables (cars, appliances).
     - Lower rates reduce the reward to saving.
  2. **Investment**
     - Lower rates make it cheaper for firms to finance new capital (machines, factories).
     - Lower rates make stocks more attractive, easing firms' equity financing.
     - Lower rates encourage residential construction.
  3. **Net exports**
     - Higher Canadian interest rates attract foreign funds.
     - The Canadian dollar appreciates, making exports more expensive and imports cheaper.
     - Net exports fall when domestic interest rates rise, and increase when they fall.

Figure 11.7 – Expansionary Monetary Policy (1 of 2)

Figure 11.7 Monetary Policy

In panel (a), short-run equilibrium is at point *A*, with real GDP of \$1.8 trillion and a price level of 113. An expansionary monetary policy causes aggregate demand to shift to the right, from AD_1 to AD_2, increasing real GDP from \$1.8 trillion to \$2.0 trillion and the price level from 113 to 115 (point *B*). With real GDP back at its potential level, the Bank of Canada can meet its goal of high employment. In panel (b), short-run equilibrium is at point *A*, with real GDP at \$2.2 trillion and the price level at 117. Because real GDP is greater than potential GDP, the economy experiences rising wages and prices. A contractionary monetary policy causes aggregate demand to shift to the left, from AD_1 to AD_2, causing real GDP to decrease from \$2.2 trillion to \$2.0 trillion and the price level to decrease from 117 to 115 (point *B*). With real GDP back at its potential level, the Bank of Canada can meet its goal of price stability.



Expansionary Monetary Policy

- **Expansionary monetary policy**:
  - BoC actions that decrease interest rates to increase real GDP.
- This is appropriate when:
  - short-run equilibrium real GDP is **below** potential GDP (recession).
- Mechanism:
  - lower interest rates $\Rightarrow$ higher consumption, investment, and net exports,
  - $\Rightarrow$ aggregate demand (AD) shifts to the **right**,
  - $\Rightarrow$ real GDP and employment increase.

Contractionary Monetary Policy

- Sometimes the economy produces **above** potential GDP:
  - unemployment is very low,
  - inflationary pressures are building.
- **Contractionary monetary policy**:
  - BoC actions that increase interest rates to reduce inflation.
- Why intentionally reduce real GDP?
  - The BoC is focused on **long-run growth**.
  - If inflation becomes a problem, tightening policy helps maintain price stability.

Can the BoC Eliminate Recessions?

- In textbook diagrams, the BoC:
  - knows exactly how far to shift AD,
  - and manages to shift AD by precisely that amount.
- In reality, monetary policy is much harder:
  - the BoC does **not** know the exact size of shocks,
  - the economy is always hit by new shocks,
  - the effects of policy occur with **lags**.
- Completely eliminating recessions is unrealistic.
- The best the BoC can do is to make recessions **milder** and **shorter**.

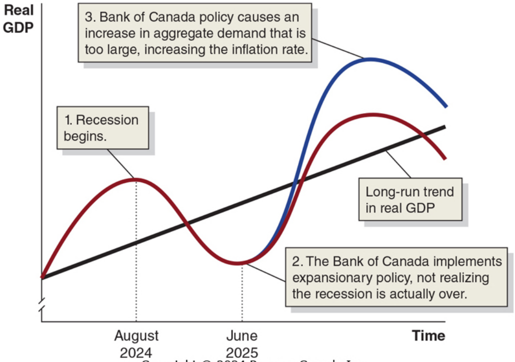
Data Lags and Policy Lags

- **Data lag**:
  - key statistics (GDP, unemployment, inflation) are available only with a delay.
  - Example: GDP for the 4th quarter of 2021 (Oct–Dec) is only published in March 2022.
- **Policy lag**:
  - it takes time for the BoC to recognize problems and implement policy changes.
- **Effect lag**:
  - it takes time for interest rate changes to affect spending, AD, and GDP.

Figure 11.8 – Poorly Timed Monetary Policy

Figure 11.8 The Effect of a Poorly Timed Monetary Policy on the Economy

The upward-sloping straight line represents the long-run growth trend in real GDP. The curved red line represents the path real GDP takes because of the business cycle. If the Bank of Canada is too late in implementing a change in monetary policy, real GDP will follow the curved blue line. As a result of the Bank of Canada's expansionary monetary policy, an increase in aggregate demand that is too large occurs during the next expansion, which causes an increase in the inflation rate.



What If Policy Is Poorly Timed?

- Suppose a recession begins in August 2020.
- Because of data and decision lags:
  - the BoC only recognizes it later.
- Assume the BoC starts expansionary policy in June 2021, when:
  - the recession has already ended and recovery is underway.
- Keeping interest rates low for too long:
  - pushes real GDP **far above** potential,
  - creates **high inflation**,
  - and makes the next recession more severe when policy eventually tightens.

Figure 11.9 – Expansionary and Contractionary Policies

Figure 11.9 Expansionary and Contractionary Monetary Policies

Panel (a) shows that when the Bank of Canada adopts an expansionary monetary policy, it sets off a series of steps that result in real GDP and the price level being higher than they would otherwise have been. Panel (b) shows that when the Bank of Canada adopts a contractionary monetary policy, it sets off a series of steps that result in real GDP and the price level being lower than they would otherwise have been.

(a) An expansionary policy

(b) A contractionary policy

Summary: Expansionary vs. Contractionary Policy

- **Expansionary policy:**
  - lowers interest rates,
  - increases C, I, and NX,
  - shifts AD right,
  - raises real GDP and employment,
  - may increase inflation.
- **Contractionary policy:**
  - raises interest rates,
  - reduces C, I, and NX,
  - shifts AD left,
  - reduces inflationary pressures,
  - may temporarily reduce real GDP and increase unemployment.

11.5 Learning Objective

Goal

Discuss the Bank of Canada's setting of monetary policy targets.

- Should the BoC target the money supply or the interest rate?
- What simple rules have economists proposed?
- How does **inflation targeting** fit into modern practice?

Should the BoC Target the Money Supply?

- In normal times, the BoC targets the **overnight rate**.
- Some economists asked: should it target the **money supply** instead?
- **Monetarists**, led by Milton Friedman, argued "yes":
  - advocate a monetary growth rule:
  - increase the money supply at about the long-run rate of real GDP growth.
- They argued:
  - active countercyclical policy might destabilize the economy,
  - a simple rule would provide more stability.

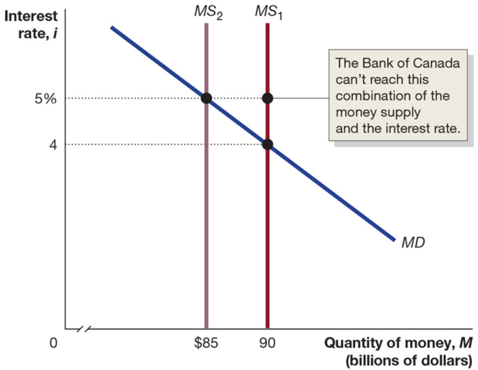
Why Monetarism Lost Influence

- Monetarism was popular in the 1970s.
- Since the 1980s, the link between:
  - the money supply and real GDP,
  - the money supply and inflation,
- has weakened.
- For example, M1+ often changes "wildly," but:
  - real GDP and inflation do not move in the same way.
- As a result, support for following a strict monetary growth rule has declined.

Figure 11.12 – Can the BoC Target Both Money and Interest?

Figure 11.12 The Bank of Canada Can't Target Both the Money Supply and the Interest Rate

The Bank of Canada is forced to choose between using either the interest rate or the money supply as its monetary policy target. In this figure, the Bank of Canada can set a target of $90 billion for the money supply or a target of 5 percent for the interest rate, but the Bank can't hit both targets because it can achieve only combinations of the interest rate and the money supply that represent equilibrium in the money market.



Why the BoC Can't Target Both M and i

- It might seem attractive to:
  - target both the money supply and the interest rate.
- But they are linked through the money demand curve:
  - decreasing the money supply raises the equilibrium interest rate,
  - increasing the money supply lowers the equilibrium interest rate.
- Therefore:
  - the BoC cannot independently fix both the money supply and the interest rate.
  - It must choose one as its primary target.

The Taylor Rule (1 of 2)

- The **Taylor rule** (John Taylor, Stanford) links the central bank's target overnight rate to macroeconomic conditions.
- A generic version:
$$i_t = r^* + \pi_t + \frac{1}{2}(\pi_t - \pi^*) + \frac{1}{2}\left(\frac{Y_t - Y_t^*}{Y_t^*}\right),$$
  where:
  - i_t: target nominal overnight rate,
  - r^*: long-run real interest rate,
  - π_t: current inflation rate,
  - π^*: target inflation rate,
  - Y_t: real GDP,
  - Y_t^*: potential GDP.

The Taylor Rule (2 of 2)

- The weights of $1/2$ on:
  - the **inflation gap** $(\pi_t - \pi^*)$,
  - and the **output gap** $\left(\frac{Y_t - Y_t^*}{Y_t^*}\right)$,
- would differ if the BoC were more or less concerned about inflation or output.
- In the Taylor rule:
  - the coefficient on the inflation gap is greater than zero,
  - so if inflation rises by 1 percentage point, the overnight rate rises by **more than** 1 percentage point.
- This is the **Taylor principle**:
  - to stabilize the economy, monetary policy must raise the real interest rate when inflation rises.
- Many economists see the Taylor rule as a useful benchmark for analyzing overnight rate decisions.

Inflation Targeting

- **Inflation targeting**:
  - a framework in which the central bank announces an explicit target for the inflation rate.
- By the 1980s, many central banks had adopted inflation targets.
- For example:
  - the BoC announces a target range for inflation,
  - the U.S. Federal Reserve publicly announced a 2% inflation target.

Why Use Inflation Targeting?

- Inflation targeting:
  - informs the public about the BoC's objectives,
  - provides an anchor for inflation expectations,
  - increases accountability for the central bank.
- Economists and policymakers debate whether inflation targets improve policy, but:
  - by 2021, many central banks had adopted inflation targeting,
  - but no major central bank had adopted **nominal GDP targeting**.

Summary

- Monetary policy: actions by the Bank of Canada to manage the money supply and interest rates in pursuit of:
  - price stability, high employment, financial stability, and economic growth.
- The BoC uses open market operations and lending to financial institutions to influence:
  - the money supply and,
  - primarily, the overnight interest rate.
- Changes in interest rates affect aggregate demand through consumption, investment, and net exports.
- Expansionary policy lowers interest rates to raise real GDP toward potential; contractionary policy raises rates to curb inflation.
- The BoC cannot target both the money supply and the interest rate simultaneously.
- Rules such as the Taylor rule and frameworks like inflation targeting help guide and communicate monetary policy decisions.

Thank you!